RAMAKRISHNA KARNATI

Azure Cloud Data Engineer | **ETL, DevOps, Visualization**

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Career Summary

Results-driven **Azure Cloud Data Engineer** with **4+ years** of experience across **Cloud Infrastructure, ETL Design, and DevOps Automation**. Specialized in building scalable Azure data platforms using **Azure Data Factory (ADF), Databricks, ADLS Gen 2, and Fabric**. Proven expertise in developing **CI/CD pipelines**, optimizing cloud costs, and delivering high-performance data solutions processing **billions of records** efficiently on daily basis. Strong track record of improving system efficiency, enabling automation, and driving business insights through **data visualization**.

Key highlights include:

- Designed and implemented end-to-end **Azure data platform architectures** leveraging **Delta Lake (Medallion architecture)**, enabling scalable, reliable, and high-performance data processing across ingestion, transformation, and consumption layers.
- Developed scalable and metadata-driven ETL pipelines in ADF migrating **terabytes** of data from various on-premise data sources to Azure, achieving a **99%** success rate.
- Built **PySpark**-based data processing solutions handling **gigabytes** of data daily across multiple formats from various sources like **SAP**, **Salesforce**, **SFTP servers**, and **on-premise databases**.
- **Led a team of 5** to implement an **LLM**-based solution to categorize **billions** of products into business provided classifications based on their descriptions and UPC codes, achieving over **90% accuracy**.
- Automated **Vacuum** and **Optimize** operations on Databricks tables, enhancing storage efficiency and boosting table performance by **40%**, and optimized query performance through improved code design and leveraging advanced techniques like **partitioning**, **broadcasting**, **caching** and **liquid clustering**.
- Architected and standardized **end-to-end data platform deployments** with integrated **CI/CD, infrastructure as code (Terraform & ARM Templates), and automated governance** using **Azure DevOps**.
- Integrated DevOps **YAML**-pipelines with **SonarQube** to automate code quality checks, reducing review effort by **80%**.
- Developed centralized monitoring and analytics solutions using **Power BI** and Databricks, enabling near real-time visibility into pipeline performance, SLA compliance, and **LLM usage metrics** across large-scale data platforms.
- Implemented robust data governance and quality frameworks using **Unity Catalog**, **Purview**, **RLS**, and validation rules integrated with ETL, ensuring secure, reliable, and high-quality data for downstream analytics.

Education

B.Tech, Electronics & Communication Engineering
Vardhaman College of Engineering, Hyderabad — *Jun 2018 – May 2022*

Skills

Programming: Python, PySpark, SQL, Java, Scala, PowerShell, Unix

Cloud Tools: Azure Data Factory, Databricks, ADLS, Logic Apps, Azure DevOps, Automation Account, Purview, Terraform, ARM Templates, Microsoft Fabric (Warehouse, Lakehouse, Data Engineering, Data Science), Power BI, AWS (EC2, S3)

Other Tools: Git, HTML, CSS, JavaScript, Spring Boot, Docker, Kubernetes, SonarQube

Soft Skills: Quick learning ability, Problem solving, Critical thinking, Attention to detail, Adaptability, Business communication

Awards & Achievements

- Recognized by internal delivery heads for representing my current project as the primary presenter during an external audit, delivering end-to-end walkthrough of architecture design, delivery processes, and documentation as per the comprehensive audit checklist, contributing to Infosys securing the Data Warehouse Migration to Microsoft Azure specialization.
- Earned an official appreciation from the client for going above and beyond in resolving a critical incident during a crucial month-end business closure period outside the scope of the project and regular work hours.
- Received a designated client appreciation for delivering exceptional cross-project support by automating CI/CD deployments and providing expertise in infrastructure automation, ETL best practices, and DevOps excellence.
- Consistently recognized with "**Rise Insta**" awards for exceptional performance every quarter and awarded with the "**Business Ninja Award**" in recognition of remarkable contributions and outstanding achievements.
- Achieved the "**Outstanding**" rating twice in year-end performance appraisals, reflecting exceptional contributions.

Certifications

- Databricks Certified: Data Engineer Professional
- Microsoft Certified: Azure Solutions Architect Expert (AZ-305)
- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)

Experience

Specialist Programmer, Infosys, Hyderabad *Aug 2022 – Present*

Cloud Infrastructure & DevOps:

- Provisioned full-stack Azure environments (ADLS, Databricks, ADF, VM) with private networking via **ARM Templates** and **Terraform**, cutting setup time to just around **20 minutes**, an **85%** reduction compared to manual configuration time.
- Designed and implemented end-to-end **CI/CD pipelines** in Azure DevOps using **YAML** and **Declarative Automation Bundles (DAB)**, enabling automated deployment of ADF pipelines, Databricks notebooks/workflows, and infrastructure components, aligned with standard **Git-based branching strategies** and release governance.
- Introduced a **selective deployment framework** within CI/CD pipeline, allowing targeted deployment of specific artifacts rather than the entire codebase, enhancing flexibility and reducing deployment time.
- Integrated **SonarQube** and **ARMTTK** code scanning tools into Azure DevOps pipelines to enforce infrastructure and code quality standards during deployment, reducing manual review effort by **80%**.
- Automated provisioning and configuration of Databricks clusters, pools, etc., along with **Microsoft Purview** data sources and scans using **REST APIs**, enhancing deployment efficiency by **70%** and streamlining data governance processes.
- Developed **PowerShell-based** automation leveraging Azure Automation Account and AZ modules to generate cost analytics and ADF/Databricks job performance reports, reducing manual effort by **95%** and enabling proactive cost optimization.

ETL & Data Engineering:

- Designed and built scalable and metadata-driven **ETL pipelines** using Azure Data Factory and Databricks, orchestrating ingestion of **terabytes of data** from SAP, Salesforce, SFTP, and RDBMS sources into Delta Lake with a **99% success rate**.
- Developed **PySpark-based** data processing frameworks to transform and load **GB–TB scale data** across multiple formats (CSV, JSON, Parquet, XML), enabling efficient batch processing for downstream analytics.
- Implemented **event-driven data ingestion architecture** using event based triggers in ADF, automating processing of **20+ daily file arrivals** and reducing manual intervention by **70%**.
- Engineered end-to-end ingestion workflows integrating **Azure Logic Apps, ADF, and Databricks** to extract SAP data via RFCs and process it into curated Delta Lake layers.
- Optimized **Delta Lake performance** by implementing **partitioning, liquid clustering, OPTIMIZE, and VACUUM**, improving query performance by **40%** and reducing storage overhead.
- Built robust **logging and alerting frameworks** in ADF with custom error handling and **HTML**-based notifications, enabling faster incident resolution and pipeline observability.
- Developed high-throughput **JDBC-based** ingestion pipelines to load **millions of records daily** from on-prem databases into Delta Lake tables through Databricks workflows.
- Led a team of 5 to Design a large-scale data classification pipeline for **2.5B+ records** using hybrid approaches (rule-based, keyword, and **LLM-based via Azure OpenAI**), achieving over **90% accuracy**.
- Engineered a **cost-optimized LLM processing strategy** by sampling **0.2% of over 2 billion records** of data to generalize classification, significantly reducing compute costs while maintaining accuracy.
- Built unified & re-usable **Databricks workflows** to process both historical and incremental data, ensuring scalable and consistent data pipelines across ingestion layers.

Reporting & Visualization:

- Designed and developed centralized **Power BI** monitoring dashboard by ingesting **ADF pipeline logs** via Databricks tables, enabling near real-time monitoring of **pipeline health, failure rates, and SLA compliance** across **150+** pipelines.
- Designed and developed **AI/BI dashboards in Databricks** to monitor **LLM usage metrics**, including input/output token consumption, request volume, and cost analysis, enabling real-time visibility into model utilization.
- Built a **data reconciliation** framework in Power BI integrating data from **Azure SQL, Databricks, and SharePoint**, to detect discrepancies across systems and trigger notifications to respective stakeholders through **Power Automate**.
- Modeled optimized **star schema semantic layers** by de-normalizing complex relational datasets into fact and dimension tables, leveraging **DAX measures, relationships, and aggregations** to improve report query performance by **30%**.
- Automated **Power BI dataset refresh orchestration** using **REST APIs** integrated with ADF pipelines, enabling near real-time data availability post ETL completion and reducing data latency from **2 hours to 10 minutes**.

Data Governance:

- Implemented centralized **data governance using Unity Catalog**, managing **external and managed Delta tables** with a unified metastore, enabling fine-grained access control, data lineage tracking, and secure data sharing across environments.
- Established **data quality monitoring frameworks** for curated Delta tables, profiling datasets to track schema consistency, data freshness, and usage patterns over time.
- Designed and enforced **Row-Level Security (RLS)** on **100M+ record datasets** using **Azure AD group-based access control**, ensuring secure and role-based data access for downstream consumers.
- Integrated **data validation rules** within ETL pipelines to enforce business constraints, detect anomalies, and prevent propagation of invalid data into curated layers.

Senior Analyst Intern, Capgemini, Hyderabad *Feb 2022 – Apr 2022*

- Collaborated within a team of 5 to develop a **Java-based Spring Boot** application for student attendance management, utilizing **PostgreSQL** and **JDBC** for efficient database operations and GitHub for version control.
- Ensured application reliability with **90%+** test coverage using **JUnit**, and deployed containerized solutions using **Docker** on **AWS EC2** with **Kubernetes** for scalable orchestration.